A K-LINEAR EQUIVALENCE OF COHERENT CATEGORIES OVER FINITE TYPE SCHEMES COMES FROM A K-ISOMORPHISM AND AN INVERTIBLE MODULE (WEAK FORM OF GABRIEL)

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate. This item is a dual-proof environment: the spec-level proof_lines override collects both proof environments (LANE_BRIEF 10.2 span recomputation); the pool candidate recorded only the first.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0GPK: sources/stacks-chow/tags.txt, source lines 20830–20830. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0GPL: sources/stacks-chow/tags.txt, source lines 20831–20831. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference shared-bibliography-key: sources/stacks-functors/SOURCE.json, source lines 14–15. Author's bibliographic citation inside the counted span (reference environment pointing at Gabriel); the shared Stacks bibliography is not staged locally, so the cite key is declared here per SOURCE.json bibliography_note. Pointer only; its content is not part of this item.

Lemma 0.1. *Let k be a field. Let X, Y be finite type schemes over k with X separated and Y reduced. If there is a k-linear equivalence $F : Coh(\mathcal{O}_X) \to Coh(\mathcal{O}_Y)$ of categories, then there is an isomorphism $f : Y \to X$ over k and an invertible $\mathcal{O}_Y$-module $\mathcal{L}$ such that $F(\mathcal{F}) = f^*\mathcal{F} \otimes \mathcal{L}$.*

Proof using Gabriel-Rosenberg reconstruction. This lemma is a weak form of the results discussed in Lemma 0GPK and Remark 0GPL. □

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